

Program 8:

Project Structure:

Pr8

->server.js

->nodejs-configmap.yaml

->nodejs-pod.yaml

Step 1: Create [Server.js](#) file

```
JS server.js > ...
1  const http = require("http");
2  const port = process.env.PORT || 3000;
3
4  const server = http.createServer((req, res) => {
5    res.end("Hello from Node.js running inside Kubernetes!");
6  });
7
8  server.listen(port, () => {
9    console.log(`Server running at port ${port}`);
10 });
```

Step 2: Create nodejs-configmap.yaml

```
! nodejs-configmap.yaml
1  apiVersion: v1
2  kind: ConfigMap
3  metadata:
4    name: nodejs-app-config
5  data:
6    server.js: |
7      const http = require("http");
8      const port = process.env.PORT || 3000;
9
10     const server = http.createServer((req, res) => {
11       res.end("Hello from Node.js running inside Kubernetes!");
12     });
13
14     server.listen(port, () => {
15       console.log(`Server running at port ${port}`);
16     });
```

Step 3: Create nodejs-pod.yaml

```
! nodejs-pod.yaml
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: program-8
5    labels:
6      app: nodejs
7  spec:
8    containers:
9      - name: nodejs
10        image: node:18
11        command: ["node", "/usr/src/app/server.js"]
12        ports:
13          - containerPort: 3000
14        volumeMounts:
15          - name: app-code
16            mountPath: /usr/src/app
17        volumes:
18          - name: app-code
19            configMap:
20              name: nodejs-app-config
```

Step 4: Apply YAML files

```
kunaltiwari@riot pr8 % kubectl delete pod program-8
kubectl apply -f nodejs-pod.yaml
```

```
pod "program-8" deleted from default namespace
pod/program-8 created
```

Step 5: Describe

```
kunaltiwari@riot pr8 % kubectl describe pod program-8
```

Name: program-8
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Mon, 01 Dec 2025 15:54:29 +0530
Labels: app=nodejs
Annotations: <none>
Status: Pending
IP: <none>
IPs: <none>

Containers:

nodejs:

Container ID: <none>
Image: node:18
Image ID: <none>
Port: 3000/TCP
Host Port: 0/TCP
Command: node
/usr/src/app/server.js
State: Waiting
Reason: ContainerCreating
Ready: False
Restart Count: 0
Environment: <none>
Mounts:
/usr/src/app from app-code (rw)
/var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-w845z (ro)

Conditions:

Type	Status
PodReadyToStartContainers	False
Initialized	True
Ready	False
ContainersReady	False
PodScheduled	True

Volumes:

app-code:

Type: ConfigMap (a volume populated by a ConfigMap)
Name: nodejs-app-config
Optional: false

kube-api-access-w845z:

Type: Projected (a volume that contains injected data from multiple sources)
TokenExpirationSeconds: 3607
ConfigMapName: kube-root-ca.crt
Optional: false
DownwardAPI: true

QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:

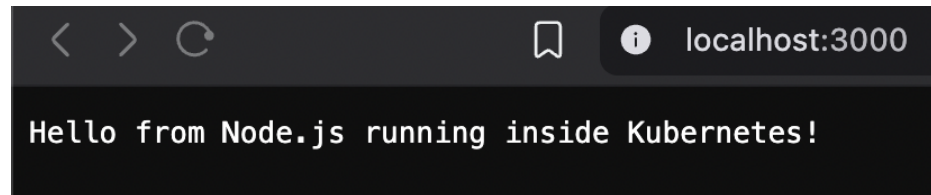
Type	Reason	Age	From	Message
Normal	Scheduled	2m29s	default-scheduler	Successfully assigned default/program-8 to minikube
Normal	Pulling	2m29s	kubelet	Pulling image "node:18"

Step 6: Access the Node.js app

```
kunaltiwari@riot pr8 % kubectl port-forward pod/program-8 3000:3000

Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
^C%
```

Step 7: open localhost:3000



Step 8: Clean up process

```
kunaltiwari@riot pr8 % kubectl delete pods -all
error: unknown shorthand flag: 'a' in -all
See 'kubectl delete --help' for usage.
kunaltiwari@riot pr8 % kubectl delete pods --all
pod "program-8" deleted from default namespace
^C%
kunaltiwari@riot pr8 % kubectl delete svc --all
service "kubernetes" deleted from default namespace
kunaltiwari@riot pr8 % kubectl delete deployments --all
No resources found
kunaltiwari@riot pr8 %
```